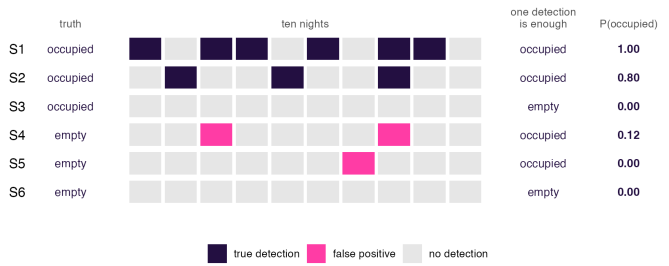
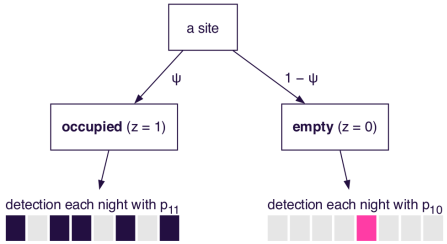
The model and the known truth

The problem



A classifier misses calls, and it also cries wolf. Counting a site as occupied after one detection inflates occupancy. crywolf lets detections be false and gives each site a probability of being occupied.

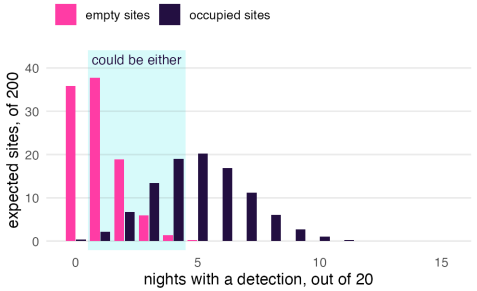
Two stories for every site



symbol	meaning	formula
ψ	a site is occupied	`occ = ~ x`
p11	detection at an occupied site	`det = ~ x`
p10	detection at an empty site	`fp = ~ x`

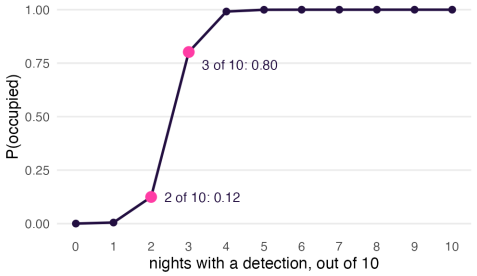
The gap Δ keeps p11 above p10 at an average site, so the two states cannot swap labels.

How the model tells them apart



Repeated occasions separate occupied from empty sites. Where the groups overlap, the data alone cannot say how many detections are false. Known truth settles that.

P(occupied) of one site

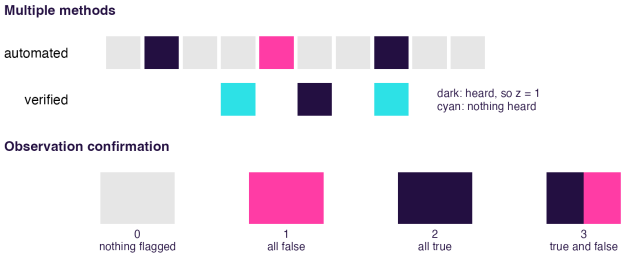


With n_1 occasions with a detection and n_0 without:

Occupied	$\psi \times p11^{n_1} \times (1 - p11)^{n_0}$
Empty	$(1 - \psi) \times p10^{n_1} \times (1 - p10)^{n_0}$
P(occupied)	Occupied ÷ (Occupied + Empty)

`fp_sites()` repeats this for every posterior draw. A verified or confirmed true detection gives P(occupied) = 1.

It needs some known truth



Without it, p10 rests only on the priors and the gap, and `fp_occu()` warns about it.

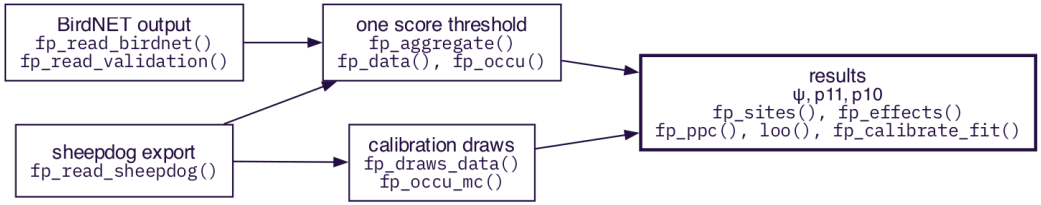
	multiple methods	observation confirmation
observes	a second method, on occasions of its own	flags on random flagged occasions
recovers best	p11	p10
anchors	ψ, the state	the error rate at the threshold
blind to	false alarms at occupied sites	where the classifier never fires
new threshold	nothing to redo	redo the checks
cost	high	low
`fp_data()`	`verified =`	`confirmed =`

Draw the checked occasions at random, at the threshold of the analysis. Clips picked for high scores understate p10.

The same verdict in `verified` and in the confirmed rows counts twice, and `fp_data()` warns about it.

From detections to a fit

Choose a route



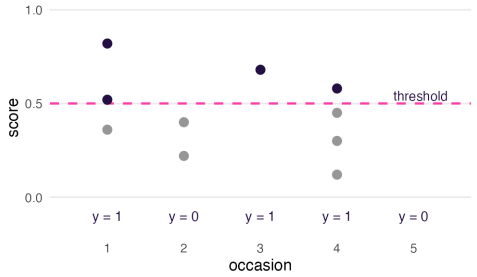
Read the data

```
# BirdNET selection tables and reviewed clips
detections <- fp_read_birdnet("tables")
verdicts <- fp_read_validation("verdicts.csv")

# or an export of sheepdog
x <- fp_read_sheepdog("occupancy_export",
                      species = "compau")
```

`fp_read_sheepdog()` drops clips drawn with `sampling_mode = "top"`, with a warning.

Build the occasions




```
auto <- fp_aggregate(occasions, detections,
  threshold = 0.5, validation = verdicts)
```

`occasions` lists every recording, including those without detections. For a sheepdog export, use `by = c("site", "occasion_id")`.

Assemble the data

```
# multiple methods
d <- fp_data(auto, sites, verified,
             recorder = "recorder")

# observation confirmation
d <- fp_data(auto, sites,
             confirmed = c("n_true", "n_false"))

# detection matrices instead of long tables
d <- fp_data_wide(y_auto, y_verified, site_covs)
```

`auto` has one row per occasion and `sites` one row per site. Add `season` for the dynamic model and `species` for several species.

Set the priors

```
fp_priors(slope_sd = 1, gap_sd = 1)
fp_gap_prior(n_true = 46, n_false = 4,
             p11 = 0.6, psi = 0.5)
pp <- fp_prior_predictive(d, occ = ~ x, fp = ~ 1)
plot(pp)
```

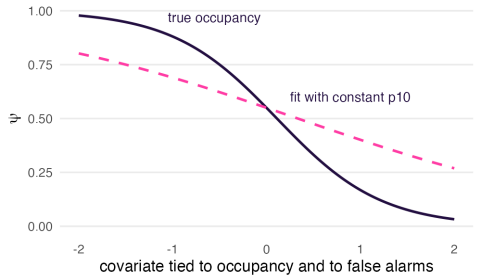
`fp_gap_prior()` turns reviewed counts into a prior for the gap. Rough values of p11 and ψ come from a pilot fit or earlier work.

Fit

```
m <- fp_occu(d, occ = ~ x, det = ~ 1, fp = ~ w,
             seed = 1)
```

argument	adds
`ver = ~ listener`	covariates of the verified method
`random_recorder`	an intercept per recorder
`random_fp`, `random_det`	a site effect on each rate
`markov = "time"`	runs of true detections
`priors =`	a call to `fp_priors()`

What can change p10



When false positives follow a gradient tied to occupancy, a constant p10 flattens the occupancy slope. Let `fp` depend on that gradient.

Check, report and extend

Check the fit first

```
summary(m)
plot(m, type = "trace")
plot(m, type = "rank")
ppc <- fp_ppc(m, seed = 1)
plot(ppc)
```

Do not read the estimates before looking at R-hat, the effective sample sizes, the divergences and the posterior predictive check. Use `fp_ppc(m, method = "verified")` for the verified occasions.

Read the results

```
coef(m, "occ")
summary(m, prob = 0.9)
as_draws_df(m)
plot(m)
plot(fp_effects(m, "x", process = "fp"))
predict(m, newdata, process = "occ")
st <- fp_sites(m, seed = 1)
plot(st)
```

`fp_sites()` gives P(occupied) and the false-positive rate of each site.

Terms

site	the unit whose occupancy is estimated
occasion	one survey of a site: a recording, a minute or a night
y	1 if the detector flagged the species on an occasion
z	the hidden state of a site: 1 occupied, 0 empty
threshold	the score from which a prediction counts as a detection
P(tp)	the probability that one detection is true

Compare models

```
l1 <- loo(m1)
l2 <- loo(m2)
fp_loo_weights(constant = l1, on_w = l2)
```

Test the threshold

```
sweep <- fp_threshold_sweep(occasions, detections,
  thresholds = c(0.1, 0.4, 0.7), sites = sites,
  verified = verified, occ = ~ x, fp = ~ 1)
plot(sweep)
```

A lower threshold lets in more true and more false detections, and p10 is estimated at that threshold.

Carry the calibration into occupancy

```
x <- fp_read_sheepdog("occupancy_export",
                      species = "nyc_alb")
versions <- fp_draws_data(x, n_draws = 20, seed = 1)
mc <- fp_occu_mc(versions, occ = ~ 1, fp = NULL)
summary(mc)
```

Each version of the detections gets one fit, and the posteriors are pooled. `sd_within` is the spread within a fit and `sd_between` the spread between fits. The occupancy model does not update the calibration (a cut model).

`fp = NULL` fits no false-positive process. To estimate p10 on this route, keep `fp` and give annotated recordings (`verified`).

What to report

model	the design, what an occasion is and how the checked occasions were chosen
priors	every prior, with a reason
computation	R-hat, effective sample sizes and divergences
adequacy	posterior predictive check and model comparison
sensitivity	refits with other priors

See `vignette("reporting")`, which has a template.

Plan and check a design

```
sim <- simulate_fp(nsite = 300, nocc = 20,
                   p10 = 0.05, frac_verified = 0.3)
fp_verification_curve(p10 = 0.05,
  conf_effect = 1.5, conf_cor = 0.7,
  models = list(constant = ~ 1, on_x = ~ x))
cal <- fp_calibrate_fit(m,
  models = list(constant = ~ 1, on_w = ~ w))
summary(cal)
```

`fp_calibrate()` scores the coverage and bias of a design before the fieldwork. `fp_calibrate_fit()` does it for data like the real ones.

Dynamic model

```
d <- fp_data(auto, sites, verified,
             season = "night", season_covs = daily)
fit <- fp_occu_dyn(d, psi1 = ~ 1, col = ~ temp,
                   ext = ~ temp, fp = ~ 1)
```

Several species

```
d <- fp_data(auto, sites, verified,
             species = "species")
ms <- fp_occu_ms(d, occ = ~ x, fp = ~ 1)
```

Rare species borrow strength from the others. There is no data augmentation for species never detected. `fp_occu_ms()` is in development.

Learn more

`vignette("crywolf")`	the model, with equations
`vignette("birdnet_workflow")`	from BirdNET to a fit
`vignette("reporting")`	what to report
Illustrating how crywolf works	the model in drawings